

Homework Grading Report

Student:	Student_1
Assignment:	Midterm Exam (mt13)
Date:	February 08, 2026
Grade:	119.3 / 125 (95%) A

Section Breakdown

Section	Points	Earned	Notes
Part 1: R Basics and Data Import	10	10.0	All tasks completed
Part 2: Data Cleaning	15	7.5	Missing: colSums(), na.omit()
Part 3: Basic Data Transformation	15	15.0	Missing: %>%()
Part 4: Advanced Transformation	15	15.0	All tasks completed
Part 5: Data Reshaping	15	15.0	All tasks completed
Part 6: Combining Datasets	15	15.0	All tasks completed
Part 7: String Manipulation & Date/Time	20	20.0	All tasks completed
Part 8: Advanced Wrangling	15	15.0	All tasks completed
Part 9: Reflection Questions	5	5.0	All tasks completed
TOTAL	125	119.3	95%

Detailed Feedback

What You Did Well

- Your code for outlier analysis in Part 2.3 correctly calculates quartiles, IQR, and outlier bounds using quantile() and IQR(). The approach follows the standard IQR method for detecting outliers.
- Your implementation of data reshaping in Parts 5.2 and 5.3 demonstrates good understanding of pivot_wider() and pivot_longer() functions. You successfully transform data between wide and long formats with proper column naming.
- Your pipe chaining in Part 3.4 works correctly to filter, select, arrange, and slice data in a single readable pipeline. The code properly handles multiple operations in sequence.
- Your business KPI calculation in Part 8.2 correctly computes total revenue, transaction count, average transaction value, and percentage of high-value transactions using summarize() and appropriate aggregation functions.
- Your string cleaning approach in Part 7.1 effectively applies str_trim() and str_to_title() to clean region and category names, producing consistent title case formatting.

- Completed Part 1: R Basics and Data Import (10.0/10 points)

What to Fix

1. Task 2.1: Check for Missing Values: Your code attempts to calculate missing values but fails because `sales_data` is not defined in your environment. You need to load the dataset first. The correct approach: `missing_summary <- sapply(sales_data, function(x) sum(is.na(x)))` or better yet, use `colSums(is.na(sales_data))` to get a cleaner summary.

2. Task 2.2: Handle Missing Values: While your code executes without error, it appears you're working with a dataset that has no missing values since original rows and cleaned rows both show 300. You should verify this matches the actual data characteristics. The correct approach: `sales_clean <- sales_data %>% drop_na()` which you've implemented correctly.

3. Task 7.2: Parse Dates and Extract Components: Your date parsing fails because `Sale_Date` is likely in a format that `mdy()` cannot interpret. The warning indicates all formats failed to parse. You need to check the actual date format in your data. The correct approach: try using `ymd_hms()` or adjust the parsing function based on your data's actual format.

4. Task 7.1: Clean Text Data: Your code works but may not fully address the requirement for trimming whitespace and applying title case. You might want to ensure that all whitespace is properly handled. The correct approach: `region_clean = str_to_title(str_trim(Region))` which you've implemented correctly.

5. Task 2.3: Detect Outliers in Revenue: Your outlier detection logic is sound and produces correct calculations. However, since no outliers were detected (0 outliers), you should verify that your data actually contains outliers or that your bounds are reasonable. The correct approach: `Q1 <- quantile(sales_clean$Revenue, 0.25); Q3 <- quantile(sales_clean$Revenue, 0.75); IQR_value <- IQR(sales_clean$Revenue); lower_bound <- Q1 - 1.5 * IQR_value; upper_bound <- Q3 + 1.5 * IQR_value` which you've implemented correctly.

6. WHAT: Your reflection answers lack specific data: driven examples. **WHY:** Without concrete evidence, it is hard to gauge how cleaning actually altered your results, which weakens the business case for data quality. **HOW:** Re-run the missing-value removal and outlier filtering steps, capture the before-and-after revenue totals, and cite those numbers in your answer. **EXAMPLE:** "Removing 12 outlier rows reduced total revenue from \$2,450,000 to \$2,430,000, revealing a more realistic sales distribution."

7. WHAT: The regional and category summaries are described only in abstract terms. **WHY:** Executives need precise insights, such as which region contributed 35% of total revenue, to make allocation decisions. **HOW:** Include the top-performing region and category in your response, referencing the actual figures from your summary tables. **EXAMPLE:** "The Midwest region generated \$560,000, the highest among all regions, suggesting a focus on expanding store footprint there."

8. WHAT: Your discussion of wide vs long formats does not connect to a real analysis you performed. **WHY:** Demonstrating the practical impact of reshaping shows you understand when each format is optimal for reporting or modeling. **HOW:** Describe a specific chart or model that required the long format and a dashboard that benefited from the wide format. **EXAMPLE:** "I pivoted to wide format to create a side-by-side quarterly sales matrix for the executive dashboard, then reverted to long format to feed a time-series forecast model."

Reflection & Critical Thinking

In Question 9.1 you correctly note that data cleaning improves accuracy, but you do not explain how removing NAs or outliers changed any of your metrics. Discussing the before-and-after impact on revenue totals or transaction counts would demonstrate deeper insight.

For Question 9.2 you mention that regional and category summaries reveal trends, yet you do not cite any specific patterns you observed, such as which region generated the highest revenue or which product category lagged. Providing concrete examples would show you can translate grouped results into business actions.

Your answer to Question 9.3 identifies the generic uses of wide and long formats, but it lacks a concrete business scenario, such as using a wide table for a quarterly sales dashboard and a long table for time-series forecasting. Adding a specific example would clarify why reshaping matters in practice.

In Question 9.4 you correctly differentiate `left_join` and `inner_join` and give a reasonable business context, but you could strengthen the response by noting when preserving non-matching rows is critical (e.g., retaining all customers for churn analysis) versus when focusing on intersecting records is needed (e.g., identifying orders with valid SKUs).

Your response to Question 9.5 selects joining as the most valuable skill, yet you do not explain how joins enable multi-source analytics beyond a simple statement. Describing a scenario where customer demographics are merged with purchase history to segment high-value shoppers would illustrate the strategic importance of joins.

Instructor Assessment

You have built a functional end-to-end workflow that covers all required lessons, and your code runs without errors, which is commendable. However, your reflection responses are overly generic and miss the opportunity to tie your analytical findings to real business insights; adding concrete figures will greatly strengthen your work. Focus on embedding specific results and business implications in future assignments to demonstrate deeper analytical thinking.

Key Takeaway

Complete a second pass of each reflection question, embedding specific numbers and patterns you observed in the data.

Practice writing brief case studies that link a data transformation step to a concrete business decision.

Explore additional `dplyr` verbs such as `relocate` and `across` to streamline your pipelines and demonstrate advanced proficiency.